

Autonomous Theory Construction Laboratory

Lecture 14: Preparing a Research Release

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Overview

A release candidate fixes scope, data, grammar, environment, and acceptance criteria before final validation.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$RC = \text{freeze}(\text{scope}, \text{grammar}, \text{registry}, \text{baseline})$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$\text{criteria} = (L_R < R_*, N_{\text{fail}} = 0, p_{\text{replay}} > p_*, C_{\text{doc}} = 1)$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$\text{bundle} < 25\text{MB}$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Release Preservation Checklist
- Yun, ch. 14
- Final Project Protocol